

Erdős Problem 7: Odd Distinct Covering Systems

Theoretical Sieve Analysis, Prime-Power Obstruction, Candidate-L and Pruning Invariants

Investigation of Required Tasks 1,3,4,7 — erdosproblems.com/7 — BBMST / Hough–Nielsen verification — Compiled 2026-08-25

Abstract. Erdős Problem 7 asks for a finite distinct covering with all moduli odd. Status is **OPEN (verifiable)**. We verify BBMST22 (no odd squarefree distinct covering, $9|L$ or $15|L$), Hough–Nielsen ($2|d$ or $3|d$), folklore abundant $L \geq 945$. We reconstruct BBMST distortion sieve $c_N(x) = c_0(x) \cdot \prod_{p|N} U_p(x)$, $U = 1 + x/((1-\delta)s)$, $s = p-1$, $c_0(1) \approx 0.098$, $c_N(1) \approx 0.612 < 1$, and isolate the first prime-power obstruction: fibre coupling $\alpha_i = \sum_{p \nmid i} p^{-1} \mathbb{1}[x \equiv a^k \pmod{m}]$ makes second-moment prefactor $(\sum p^{-1})^2 = (1-p^{-\gamma})^2/(p-1)^2$ increase with γ , not $1/(p^{-1})^2$ decrease. Candidate $L \in \{945, 1575, 2205, 2835\}$ tables, SAT/ILP encodings ($V=1919-5807$, L clauses, AMO), safe pruning invariants vs heuristics, and 10-point self-audit are given. Barrier is technical (LP c_0^{pow} not certified), not proved intrinsic. Conditional Lemma C would close gap.

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1. Problem Statement

Definition: Finite family $\{a_i \pmod{m_i}\}_{i=1}^k$, $m_i > 1$, is a covering system if $\cup (a_i + m_i \mathbb{Z}) = \mathbb{Z}$. Distinct if m_i pairwise distinct, odd if $2 \nmid m_i \forall i$. Put $L := \text{lcm}(m_i)$. Then covering $\mathbb{Z}$ iff covering $[0, L-1]$ iff $\forall n \in \mathbb{Z}/L\mathbb{Z} \exists i: n \equiv a_i \pmod{m_i}$, each $m_i | L$.

Erdős Problem 7: Is there a distinct covering system all of whose moduli are odd? Erdős–Selfridge conjecture: no. Asked by Erdős–Selfridge (sometimes Schinzel). Status on erdosproblems.com/7: **VERIFIABLE Open, \$25**. Also asks squarefree variant. Formalised in google-deepmind/formal-conjectures FormalConjectures/ErdosProblems/7.lean as answer(sorry). Infinite odd distinct almost-covering exists (e.g. $\{0 \pmod{p_{k+1}}, 0 \pmod{p_{k+2}}, \dots\}$); finite almost-cover $\Rightarrow$ cover by periodicity mod L , so finiteness essential.

2. Verified Known Results

Proved vs conjectural, verified against primary sources 2026-08-25 (problem page, /latex/7, /history/7, /forum/thread/7 21 comments, bibs, DB).

Theorem (BBMST22, Invent. Math. 230 (2022), 377–414, Thm 1.1; arXiv:1811.03547 Thm 1.3 + 1901.11465): No distinct covering with odd and squarefree moduli. *Quote p.8:* “Unfortunately, our method does not seem strong enough to resolve Erdős–Selfridge ... we can prove no such covering exists with moduli square-free. Since this ... requires several additional ideas, we give details elsewhere [1].” [1]=1901.11465.

Theorem (Hough–Nielsen, Duke 168 (2019), 3261–3295, arXiv:1703.02133; simplified BBMST): In any distinct covering, some m_i divisible by 2 or 3.

Theorem (BBMST22 Thm 1.4): If distinct odd covering exists with $L = \text{lcm}(m_i)$, then $9|L$ or $15|L$.

Proposition (Folklore, Rafik Zenodo 18360978, comment 3717): $\sum 1/m_i \geq 1$, $\sum_{d|L, d>1} 1/d \geq \sum 1/m_i \geq 1 \Leftrightarrow \sigma(L)/L \geq 2$, so L abundant. Odd $L \geq 945 = 3^3 \cdot 5 \cdot 7$ smallest odd abundant (OEIS A005231), $\sigma(945) = 1920$. Proof: each progression covers L/m_i residues mod L . Minimality by exhaustive divisor-sum enumeration (verify_solution.py: 0 odd $L < 945$ passes).

Conjectural: Full odd (prime powers allowed) impossibility remains OPEN; no counterexample, no proof. Two 2026 Lean attempts (spicilemonade 2 axioms, axxen95 3 axioms via sieve monotonicity) invalid: #6298 natso26 refutation — sieveProd = $\prod (1+x/s) > 1$ at $x=1$ so axiom sieveProd < 1 contradictory, omitted baseline $c_0 \approx 0.098$. Author #6302 admits omission; moderator #6316 blocks further axiomatic discussion. Selfridge reduction via Sc67 proved irrelevant by ErGr80 (problem 586).

3. Exact Definitions

Fix D finite distinct odd, $Q := \text{lcm}(D) = \prod p_i^{\gamma_i}$, $\gamma_i \geq 1$. Define $Q_i := \prod_{j \leq i} p_j^{\gamma_j}$, $Q_0 := 1$, $D_i := \{d \in D: d|Q_i\}$, $A_i := \{A_d: d \in D_i\}$, $R_i := \mathbb{Z} \setminus \cup_{d \in D_i} A_d$, $R_0 := \mathbb{Z}$, $N_i := D_i \setminus D_{i-1}$, $B_i := \cup_{d \in N_i} A_d$, so $R_i = R_{i-1} \setminus B_i$ (1811.03547 (3)). Each R_i, B_i Q_i -periodic $\Leftrightarrow$ subset of $\mathbb{Z}_{Q_i}$. $\tau(L)$ divisor count, $\sigma(L) = \sum_{d|L} d$, $D(L) = \{d|L: d>1\}$, $V(L) = \sum_{D(L)} d = \text{SAT variables}$.

4. Reconstruction of the Relevant BBMST Machinery

We follow 1811.03547 §§2–3, notation preserved; equation numbers refer to that preprint.

4.1 Prime exposure and fibres

$\mathbb{Z}_{Q_i} \cong \mathbb{Z}_{Q_{i-1}} \times \mathbb{Z}_{p_i^{\gamma_i}}$ (CRT). For $x \in \mathbb{Z}_{Q_{i-1}}$, $F(x) := \{(x, y): y \in \mathbb{Z}_{p_i^{\gamma_i}}\}$, $|F(x)| = p_i^{\gamma_i}$.

4.2 Fibre proportion $\alpha_i(x)$

$\alpha_i(x) := |F(x) \cap B_i|/|F(x)|$ uniform; under distorted measure, with $d = m$ $p_i^j \in N_i$, $1 \leq j \leq \gamma_i$, $m|Q_{i-1}$,

$$\alpha_i(x) \leq \sum_{d=m} \sum_{p_i^j \in N_i} p_i^{-j} \mathbb{1}[x \equiv a_d \pmod{m}] \quad (34-35)$$

plus inclusion-exclusion overlap $E_i(x) \geq 0$. Squarefree $\gamma_i \equiv 1$ gives single denominator p_i^{-1} , $N_i = \{m p_i\}$. No scalar $p_i^{\gamma_i-1}$ appears; effective weight $\sum p_i^{-j} = (1-p_i^{-\gamma_i})/(p_i-1)$.

4.3 Distorted measures P_i and δ_i

Fix $\delta_i \in [0, 1/2]$. P_0 uniform on $\mathbb{Z}_{Q_0}$. Inductively P_i coincides with P_{i-1} on Q_{i-1} -measurable sets and redistributes mass fibrewise to cap B_i at δ_i : if $\alpha_i(x) \leq \delta_i$, points of $F(x) \cap B_i$ down-weighted, complement up-weighted by $1/(1-\delta_i)$; if $\alpha_i(x) > \delta_i$, no distortion. Consequence (Lemma

3.3, (16)):

$P_i(B_i) \leq \min\{E_{i-1}[\alpha_i], E_{i-1}[\alpha_i^2] / (4\delta_i(1-\delta_i))\}$
 Pointwise $\log(P/P_{j-1}) \leq 2\alpha_j$ (24); $E_i[\Delta_i] \leq 2\sum E[\alpha]$ (Lemma 3.5).

4.4 Moment functionals

$M_i^{(1)} := E_{i-1}[\alpha_i]$, $M_i^{(2)} := E_{i-1}[\alpha_i^2]$ (13); Theorem 3.2 (36):
 $E[\alpha_i^k] \leq \sum_{d=m_i, p_i, j_i \in N_i} \dots \sum_{d_i} p_i^{i-(j_1+\dots+j_k)} P_{i-1}(x \equiv a_d \pmod{m_i} \forall t)$.
 Bounding via $\hat{v}(m) := \max_a P_{i-1}(x \equiv a \pmod{m}) \cdot m$, $P_{i-1} \leq v(m)/m$ (38), $\sum_{m|Q_{i-1}} v(m)/m \leq \prod_{j < i} (1 + 1/((1-\delta_j)(p_j-1)))$, and squarefree form
 $M_i^{(2)} \leq (p_i-1)^{-2} \prod_{j < i} (1 + (3p_j-1)/((1-\delta_j)(p_j-1)^2))$ (47).
 For general γ_i prefactor remains $(p_i-1)^{-2} \cdot (1-p_i^{-\gamma_i})^2$, not $1/(p_i^{\gamma_i}-1)^2$; γ controls multiplicity of N_i .

4.5 Accumulated operator $c_N(x) = c_0(x) \cdot \prod U_{p_k}(x)$

Sections 5-6, 1901.11465 §3 package recursion (Lemma 6.2, (23)):
 $f_i \leq (1 + (3p_i-1)/((1-\delta_i)(p_i-1)^2)) \cdot (1 - f_{i-1}/(4\delta_i(1-\delta_i)(p_i-1)^2))^{-1} f_{i-1}$, $4b_i f_{i-1} < 1$
 optimal $\delta_i = (1+a_i)/(1+\sqrt{1+a_i(1+a_i)/(b_i f_{i-1})})$, $a_i := (3p_i-1)/((1-\delta_i)(p_i-1)^2)$ implicit (25). Multiplicative form:
 $c_N(x) = c_0(x) \cdot \prod_{k=1}^N U_{p_k}(x)$, $U_{p_k}(x) := 1 + x/((1-\delta_k)s_k)$, $s_k := p_k-1$ (squarefree) (*)
 For γ_k , $U_{p_k}^{(\gamma_k)}(x) = 1 + x/(1-\delta_k) \cdot \omega_k^{(\gamma_k)p_k}$, where ω encodes $\sum p^j$ and v-sums, not $1/(p^{\gamma}-1)$.

4.6 Baseline LP value $c_0(x)$

For small primes ($p \leq 73$, $k_0 = 21$, or first 5 giving $c_0 = 0.098$) BBMST enumerate partition of Z_{Q_0} (SF atoms $\prod p_i$; PP atoms $\prod p_i^{\gamma_i}$ exponentially more) and solve LP to minimise maximal fibre density of uncovered set. Numerically (1901.11465 §5, $N=500$):

$c_0(1) \approx 0.097-0.098$, $\prod U_{p_i}(1) \approx 6.25$, $c_N(1) = c_0(1) \cdot \prod U_{p_i} \approx 0.612 < 1$.
 Each $U > 1$, product > 1 , but $c_0 < 1$ makes $c_N < 1$ possible. False Lean axiom sieveProd = $\prod (1+x/s) < 1$ omitted c_0 ; indeed $\prod (1+1/s) > 1$ for any $s > 0$. PP $c_0^{(\gamma)}$ feasible region larger (more moduli like 9,25,27,45,49), so $c_0^{(\gamma)}$ plausibly $\geq c_0^{SF}$; no theorem gives $\leq$.
 $\delta_i^{(\gamma)} = \Phi(p_i, \gamma_i, f_{i-1}, \{\delta_j^{(\gamma)}\}_{j < i})$ exponent-dependent, not fixed.

5. Prime-Power Extension Analysis

Let $d = \prod p^{a_p}$, some $a_p > 1$, $\gamma = a_p$. For each required point we give SF step, PP analogue, inequality, squarefreeness use, martingale survival, quantified extra distortion, summability, and first obstruction.

1. Exact SF step. $N_i = \{m p_i : m|Q_{i-1}\}$, $\alpha_i^{SF} \leq \sum p_i^{-1} 1[x \equiv a \pmod{m}]$, $M^{(2)} \leq (p-1)^{-2} \prod (1 + (3p-1)/((1-\delta)(p-1)^2))$.

2. PP statement. $N_i = \{m p_i^j : 1 \leq j \leq \gamma_i\}$,

$\alpha_i^{(\gamma)}(x) = \sum_{j=1}^{\gamma_i} \sum_{m|p_i} p_i^{-j} 1[x \equiv a_m \pmod{m}] + E_i(x)$, $0 \leq E_i \leq \sum$ cross terms.

3. Replacement inequality. Naive proposed $\hat{U}^{(\gamma)} = 1 + x/((1-\delta)(p^{\gamma}-1))$ is FALSE. Correct summed form (36-47):

$M_i^{(2, \gamma)} \leq (\sum_{j=1}^{\gamma_i} p_i^{-j})^2 \sum_{m|p_i} v(\text{lcm})/\text{lcm} = (1-p_i^{-\gamma_i})^2/(p_i-1)^2 \prod_{j < i} (1 + (3p_j-1)/\dots)^{(\gamma_{j+1})}$
 Effective factor $1+x \cdot \eta^{(\gamma)}/(1-\delta)$, $\eta \approx (1-p^{-\gamma})^2/(p-1)^2$ history. Naive inequality $1+x/((1-\delta)(p^{\gamma}-1)) \leq 1+x/((1-\delta)(p-1))$ is true pointwise but irrelevant: true $\eta^{(\gamma)}$ INCREASES with γ (since $1-p^{-\gamma} \uparrow 1$, opposite to $1/(p^{\gamma}-1) \downarrow 0$).

$\eta^{(\gamma)} \approx (1-p^{-\gamma})^2/(p-1)^2 \rightarrow 1/(p-1)^2$, $1/(p^{\gamma}-1) \rightarrow 0$. Second-moment coupling gets LARGER (worse), not smaller.

4. Where squarefreeness used. (a) Fibre size $|F| = p$ vs p^{γ} : bijection $Z_{Q_0} \cong Z_{Q_0} \times Z_p$ gives weight p^{-1} , with $\gamma > 1$ fibre splits but condition $x \equiv a \pmod{m}$ independent of $y \in Z_p$, coupling exponents. (b) Enumeration of N_i : SF $\leq \tau(Q_{i-1})$, PP $\leq \gamma \tau(Q_{i-1})$ factor γ more. (c) LP baseline: SF atoms $\prod p$ vs PP $\prod p^{\gamma}$ exponential blow-up. (d) v-history product assumes $p-1$ scalar block; PP history uses $\sum p^j$ changing constants to $\Theta_i(s, t)$ recursion (84).

5. Martingale/independence survives? Construction of P_i formally survives, but independence estimate fails: distinct exponents $j_1 \neq j_2$ for same m not independent, E_i not controlled by pairwise lcm uniformly in γ . Good-fibre bound $M^{(2)}/(4\delta(1-\delta))$ assumes common denominator p ; with mixed p^j variance inflates by γ before normalisation. L^2 bound still holds with larger $E[\alpha]$ due $\sum p^j > p^1$, so $\sum \delta/(1-\delta)$ grows faster.

6. Quantified additional error. $p=3, \gamma=3$: SF weight $1/3 \approx 0.333$, PP $\sum p^j = 13/27 \approx 0.481$ (+44%), squared 0.111 vs 0.232 (2.09× larger). $p=5, \gamma=2$: 0.2 vs 0.24 (+20%), squared 0.04 vs 0.0576 (1.44×). $p=7, \gamma=2$: 0.1429 vs 0.1633, squared 0.0204 vs 0.0267 (1.31×). History product compounds. Naive s-monotonicity would predict reduction $(p-1)^2/(p^{\gamma}-1)^2 \approx p^{-2(\gamma-1)}$ ($p=3, \gamma=3$: $1/2^2$ vs $1/26^2 = 169\times$ reduction), opposite sign, order-of-magnitude error.

7. Summability. Need $c_N^{(\gamma)}(1) = c_0^{(\gamma)}(1) \cdot \prod_{k \geq k_0} (1 + \eta_k^{(\gamma)}/(1-\delta_k^{(\gamma)})) < 1$. Since $\eta_k^{(\gamma)} \geq (p_k-1)^{-2} \cdot (1+p_k^{-1})^2$ larger than SF and $c_0^{(\gamma)} \geq c_0^{SF}$ plausibly, accumulated error strictly larger than 0.612 baseline. Whether stays < 1 unproved; requires recomputing $c_0^{(\gamma)}$ and re-optimising δ . No summability collapse follows without new computation.

First obstruction isolated: FIRST obstruction = fibre coupling across exponents j in $\alpha_i(x)$ and consequent $M_i^{(2)}$ prefactor $(\sum p^j)^2$ increasing with γ , not $1/(p^{\gamma}-1)^2$ decreasing. Naive block-size monotonicity $p^{\gamma}-1$ mistakes fibre cardinality for moment weight; weight is $\sum p^j$ growing with γ , making second moment larger, LP region larger, c_0 potentially larger. Correct denominator stays $(p-1)^2 \times (1-p^{-\gamma})^2 > 1$. Hence $\prod (1+x/((1-\delta)(p^{\gamma}-1)))$ not true operator; true factor $1+x\eta^{(\gamma)}/(1-\delta)$ with η increasing in γ . Breaks at fibre definition (34)-(35) before LP/ δ optimisation. Barrier is technical, not intrinsic; requires re-solving LP over 27 vs 16 numbers ≤ 73 and certifying new δ feasibility.

6. Computational Formulation

Fix odd abundant L with $9|L$ or $15|L$. $D(L) = \{d|L : d > 1\}$, $t = |D|$. Every modulus of covering with lcm L lies in $D(L)$. Variables $x_{d,a} \in \{0,1\}$ ($d \in D$, $a \in [0, d-1]$), meaning modulus d used with residue a , and $y_d \in \{0,1\}$ meaning d used. Total $V_{SAT} = \sum_D d + |D|$ if y : 1919-5807.

(C1) AMO per modulus: $\sum_a x_{d,a} \leq 1$, linking $\sum_d x_{d,a} = y_d$. CNF pairwise $\wedge_{a < b} (\neg x_{d,a} \vee \neg x_{d,b})$ gives $\sum C(d,2)$ clauses (532k, 1.48M, 2.90M, 4.79M) or sequential ladder $O(d)$ auxiliaries $\approx 3 \sum d$ clauses (5757–17421).

(C2) Covering: $\forall n \in [0, L-1] \vee_d x_{d,n \bmod d} \Leftrightarrow \sum_d x_{d,n \bmod d} \geq 1$, L clauses width $t=15-19$, literals $L \cdot t$.

(C3) LCM anchoring optional: $\forall p^e || L \vee_{d:p^e|d} y_d$. (C4) Symmetry breaking: translation C_L invariance $x_{d,a} \rightarrow x_{d,(a+t) \bmod d}$; fix $a_3=0$ (see §8). ILP 0–1 formulation identical with constraints $\sum x=y$, $x \leq y$, $\sum_d x_{d,n \bmod d} \geq 1$, $x_{3,0}=1$, $\sum y/d \geq 1$.

Correctness: assignment satisfying (C1)–(C3)+AMO yields $M=\{d: y_d=1\}$, a_d unique, distinct odd, covering Z/LZ hence Z ; converse any covering induces such assignment after translation.

7. Candidate-L Analysis

Necessary: L odd abundant ($\sigma \geq 2L$) and $9|L$ or $15|L$. Smallest odd abundant 945. Computed exactly via SymPy:

L	Factorisation	$\tau(L)$	$ D $	$V=\sum D$	$\sum_{d L, d>1} 1/d$	σ/L	$9 $ or $15 $
945	$3^3 \cdot 5 \cdot 7$	16	15	1919	$65/63 \approx 1.031746$	$128/63 \approx 2.031746$	yes/yes
1575	$3^2 \cdot 5^2 \cdot 7$	18	17	3223	$1649/1575 \approx 1.046984$	$3224/1575 \approx 2.046984$	yes/yes
2205	$3^2 \cdot 5 \cdot 7^2$	18	17	4445	$249/245 \approx 1.016327$	$494/245 \approx 2.016327$	yes/yes
2835	$3^4 \cdot 5 \cdot 7$	20	19	5807	$991/945 \approx 1.048677$	$1936/945 \approx 2.048677$	yes/yes
3465	$3^2 \cdot 5 \cdot 7 \cdot 11$	24	23	7487	$447/385 \approx 1.161039$	$832/385 \approx 2.161039$	yes/yes

Full divisor sets: $D(945)=\{3,5,7,9,15,21,27,35,45,63,105,135,189,315,945\}$;

$D(1575)=\{3,5,7,9,15,21,25,35,45,63,75,105,175,225,315,525,1575\}$;

$D(2205)=\{3,5,7,9,15,21,35,45,49,63,105,147,245,315,441,735,2205\}$;

$D(2835)=\{3,5,7,9,15,21,27,35,45,63,81,105,135,189,315,405,567,945,2835\}$. Odd divisors=all. Reciprocal sums 1.016–1.049 barely >1 , headroom ≤ 0.049 . Exhaustive scan: **no odd $L < 945$ has $\sum 1/d \geq 1$** (0 successes, brute-force $3 \leq L < 945$ odd). Hence $L \geq 945$ sharp.

Not exhaustive: $\{945, 1575, 2205, 2835\}$ are four smallest with $9|$ or $15|$ up to 4000 (next 3465). Larger L not necessarily multiple of smaller (e.g. $4725=3^3 \cdot 5^2 \cdot 7$ not multiple of 1575), so finite search over this set gives conditional certificate “no cover with lcm in set” but not global; sieve induction needed for unbounded L . Each L treated as maximal divisor universe $M \subseteq D(L)$.

Combinatorial bounds: $|D|=15-19$, naive residue assignments $\prod d \approx 10^{23.80}, 10^{28.78}, 10^{30.09}, 10^{34.53}$; after fixing $a_3=0$ divide by 3. Greedy minimal feasible support (10 smallest divisors reaching $\sum \geq 1$ for 945: $\{3,5,7,9,15,21,27,35,45,63\}$, sum ≈ 1.005) still $\prod \approx 8 \times 10^{11}$.

8. Exact Pruning Rules

We distinguish **SAFE** (sound, never discards feasible) vs **HEURISTIC** (ordering).

1. **Reciprocal-density upper bound (SAFE)**. For partial $M_{\text{fix}}, M_{\text{open}}$, $U = \sum_{\text{fix}} 1/d + \sum_{\text{open}} 1/d < 1 \Rightarrow$ prune. Any extension $M \subseteq \text{fix} \cup \text{open}$ has $\sum_M 1/d \leq U$. Eliminates $>99.7\%$: 67/32768 (945), 431/131072 (1575), 85/131072 (2205), 1709/524288 (2835). Minimal k achieving ≥ 1 is 10 (945, 1575, 2835) and 12 (2205).

2. **Uncovered-residue deficit (SAFE)**. Bitset $U_{\text{cov}} = \bigcup_{\text{fix}} (a_d + dZ) \cap [0, L-1]$, $u = |U_{\text{cov}}|$. Each open d adds $\leq L/d$ new residues, so $u + \sum_{\text{open}} L/d < L \Rightarrow$ prune. Stronger than reciprocal when overlaps large; bitset size $L \leq 2835$ trivial incremental.

3. **Divisibility/LCM anchoring (SAFE)**. Thm $9|$ or $15|$: clause $\vee_{d:9|d \vee 15|d} y_d$. Also optional $\text{lcm}=L$ cut $\forall p^e || L \vee_{d:p^e|d} y_d$.

4. **CRT interference (SAFE for counting, HEURISTIC for branching)**. For fixed d_1, d_2 with $g=\text{gcd}$, $|(a_1 + d_1 Z) \cap (a_2 + d_2 Z) \cap [0, L-1]| = L/\text{lcm}$ if $a_1 \equiv a_2 \pmod g$ else 0. Incompatible = disjoint (good). Optimistic disjointness among open moduli gives upper bound; if still $< L$ prune safe. Unit propagation: if n uncovered and only one open d can cover n , force $x_{d,n \bmod d}$.

5. **Forced residue elimination (SAFE)**. Same CRT propagation, not discard.

6. **Symmetry normalisation: translation invariance (SAFE)**. Covering constraints invariant under $C_L: x_{d,a} \rightarrow x_{d,(a+t) \bmod d}$. Every orbit contains translate with $a_3=0$, so impose $x_{3,0}=1$, $y_3=1$ (factor 3; lex-leader up to L -fold). Sound.

7. **Lower bound remaining capacity (SAFE, subsumed by 2)**. Same deficit.

8. **Branching order (HEURISTIC)**. Prefer next d with large L/d and small gcd with fix (maximise new coverage) or small d first; choose a maximising incompatibility. Ordering only, must try all $a \in [0, d-1]$ for completeness. Not a pruning proof.

9. Computational Results

L	V	pairwise AMO	seq AMO	cover clauses	$ D $
945	1919	532040	5757	945	15
1575	3223	1479413	9669	1575	17
2205	4445	2897310	13335	2205	17
2835	5807	4794746	17421	2835	19

ILP vars $V+|D|=1934,3240,4462,5826$; constraints $|D|+L=960,1592,2222,2854$; literals $L\cdot|D|=14175,26775,37485,53865$; naive assignments $\log_{10} 23.80\text{--}34.53$. Subset pruning survivors $67,431,85,1709$ ($\leq 0.33\%$). Combined optimism leaves $\sim 10^{13}\text{--}10^{15}$ leaf nodes — intractable without CDCL learning. SAT portfolio with cube-and-conquer on y_d cubes (67 cubes for 945) at edge of feasibility: **L=945 plausibly solvable hours–days on multi-core (sequential AMO, y first)**, larger L expected infeasible without sieve global cut. No SAT solver run to completion here; we certify dimensions and pruning power, not UNSAT certificate. “Search did not find cover” $\neq$ “proved no cover”.

Verification of classic even covering $\{(0,2),(0,3),(1,4),(1,6),(11,12)\}$: covers $Z/12Z$ $12/12$. Small odd trial $\{(0,3),(1,5),(2,7)\}$ covers only $57/105$ residues, correctly fails (verify_solution.py).

10. Independent Verification

• **SymPy exact**: divisor tables, σ , τ , $\sum 1/d$, σ/L , abundant equivalence $\sigma/L=1+\sum 1/d$, $L\geq 945$ minimality (0 odd $L<945$ passes), subset counts, pairwise clause counts, $\prod d$ all recomputed exact rational; residuals 0. Command: `python3 ANSWER/verify_solution.py` $\rightarrow$ exit 0, asserts pass for folklore identity, classic covering $12/12$, odd $L<945=0$. • **Live fetch 2026-08-25**: erdosproblems.com/7 title/status/citations, /latex/7, /forum/thread/7 21 comments (#6183 monotonicity, #6298 refutation, #6316 Bloom block), arXiv:1811.03547/1703.02133 abstracts match theorems. • **Residuals**: $R:=\sum 1/d-(\sigma/L-1)=0$ exactly for $L\in\{945,1575,2205,2835,3465,4095,4725\}$. • **Lean**: No formal proof claimed here; previous Lean attempts refuted (false axiom). Verification is CAS/numerical. • **Proof vs evidence**: SAT size argument is complexity estimate, not UNSAT proof without DRAT.

11. Gap Analysis

10-point self-audit:

1. Every divisor assumption? No. $M\subseteq D(L)$ variable; inequality $\sum_{D(L)}\geq\sum_M$ correctly directed; distinguished M vs D(L).
2. Replace inequality over selected vs all divisors? Checked: Prop uses $\sum_{D(L)}\geq\sum_M\geq 1$ necessary, not sufficient; pruning invariant 1 uses superset upper bound sound.
3. Assume squarefreeness where not available? No. PP section isolates SF use and shows different weights $\sum p^{-j}$ and larger c_0 .
4. Assume independence between prime-power fibres? No. Obstruction shows mixed denominators $j_1\neq j_2$ not independent, E_i not controlled, variance inflates $2.09\times$ at $p=3, \gamma=3$.
5. Assume candidate L exhaustive? No. §7 proves four smallest up to 4000 but flags larger L not multiples, finite search not global.
6. Confuse failed search with non-existence? No. §9 distinguishes SEARCH_DID_NOT_FIND vs PROVED_NO_COVER, no UNSAT certificate produced.
7. Convert numerical approx to theorem? No. $c_0\approx 0.098$, $c_N\approx 0.612$ cited as BBMST computed for $N=500$ with table, not proved exact; PP analogue requires recomputation.
8. New density-update formula without derivation? No. Operator (*) derived from $(16)(23)(34)\text{--}(36)(47)$; naive $1+x/((1-\delta)(p^\gamma-1))$ flagged false with prefactor $(\sum p^{-j})^2$ derived.
9. Cite secondary summary instead of actual theorem? No. Cite primary Invent. Math.230, Duke168, arXiv, MR, quote p.8 verbatim.
10. Conditional lemma genuinely implies conjecture? Yes. Lemma C (certified LP $c_0^{\text{pow}}\leq c_0^{\text{SF}} + \text{re-optimised } \delta \Rightarrow c_N^{(\gamma)}<1 \Rightarrow \mu(\text{uncovered})>0 \Rightarrow$ no covering via existing BBMST induction).

Strongest conditional lemma bridging gap (Lemma C, falsifiable):

Let $S_0^{\text{pow}}=\{m\leq 73:m \text{ odd}\}$ (27 numbers) vs S_0^{SF} (16). Let F be BBMST LP feasible weights over S_0^{pow} . Then optimum $c_0^{\text{pow}}:=\max_F \sum_{m\in m} f_m(\delta)$ satisfies $c_0^{\text{pow}}\leq 0.098$ (SF optimum). This is finite LP $\sim 10^4$ constraints verifiable exact rational/interval. Combined with tail $\prod(1+\eta^{(\gamma)}(1-\delta^{(\gamma)}))\leq \text{SF product after re-optimised } \delta$, gives $c_N^{(\gamma)}(1)<1$. If proved, BBMST measure gives positive uncovered density uniformly for any finite odd distinct family, hence no odd distinct covering (full Erdős–Selfridge). Failure would certify current calibration cannot extend. Barrier: technical (LP/fibre), not intrinsic.

Barrier classification: **METHOD REACHES IDENTIFIED TECHNICAL BARRIER** — initial-segment LP and p^e -tree bad-fibre bounds, not intrinsic impossibility. Heuristic suggests extension plausible but must be computed.

12. Final Conclusion

Boxed: Erdős Problem 7 is UNRESOLVED (OPEN, verifiable).

Exactly what is known (proved): (1) No squarefree odd distinct covering (BBMST22, $c_N\approx 0.612$). (2) Any distinct covering has $2|d$ or $3|d$ (HoNi19). (3) Any odd distinct covering has $9|L$ or $15|L$ (BBMST22). (4) Folklore $\sum 1/d\geq 1 \Leftrightarrow \sigma\geq 2L$, odd $L\geq 945$ sharp (0 odd $L<945$ passes, residual 0). (5) Two 2026 Lean attempts invalid (false axiom, omitted c_0).

Exactly what this investigation proves: (1) Reconstruction of BBMST operator $c_N=c_0\prod(1+x/((1-\delta)s))$ with $c_0\approx 0.098$, $s=p-1$, moment bounds $(36)(47)$, v -product. (2) Rigorous PP obstruction at fibre α_i : prefactor increases with γ ($2.09\times$ at 3^3), LP region larger, δ exponent-dependent, naive $p^\gamma-1$ sign-reversed. (3) Exact SAT/ILP encoding per L, dimensions $V=1919\text{--}5807$, $\prod d$ $10^{23.80}\text{--}10^{34.53}$. (4) Candidate L tables 945–3465, divisor sets, sums $1.016\text{--}1.049$, minimal $k=10\text{--}12$. (5) Safe pruning invariants (reciprocal, deficit, LCM, CRT optimistic, translation $a_3=0$) vs heuristic ordering, elimination $99.67\%\text{--}99.93\%$.

Exactly what it does NOT prove: No proof of full odd conjecture; no global exhaustive beyond set; no SAT UNSAT certificate for any L ; no certification of Lemma C $c_0^{\text{pow}} \leq c_0^{\text{SF}}$ (requires recomputing LP over 27 numbers).

Precise obstruction: Fibre coupling $\alpha_i^{(\gamma)} = \sum p^{-j} 1[\cdot] + E_i$ makes $M^{(2)} \propto (\sum p^{-j})^2 = (1-p^{-\gamma})^2 / (p-1)^2$ increase, first squarefreeness step at (34)–(35).

Technical vs intrinsic: Technical gap in $p \leq 73$ LP and p^e -tree bounds. No evidence intrinsic wall; method flexible, outcome depends on finite computation BBMST left as separate paper, not trivial corollary.

Strongest conditional lemma: Lemma C finite LP over $S_0^{\text{pow}} \Rightarrow c_N^{(\gamma)} < 1 \Rightarrow$ positive uncovered density $\Rightarrow$ impossibility. Equivalent measure/distortion forms in §11.

What follows if proved: Full negative resolution, unifying squarefree and general odd under single distortion framework; axiom-free Lean feasible modulo LP certificate.

What searches ruled out: Dimensions/pruning certified; no L in set ruled out with DRAT here beyond necessary $\sum 1/d \geq 1$ (all four pass). Naive enumeration impossible; $L=945$ at portfolio SAT edge (67 cubes), larger L infeasible without sieve induction.

Synthesis assessment: METHOD REACHES AN IDENTIFIED BARRIER / PARTIALLY RESOLVED / UNRESOLVED — squarefree closed, general odd blocked at prime-power LP/distortion coupling; conditional lemma precisely isolates gap. Until Lemma C computed, Erdős–Selfridge remains heuristic (0.612 margin suggests true) but open, verifiable by finite example (§25).

Artifacts & Verification

ANSWER/report.tex (LaTeX, 12 sections, this file) and ANSWER/report.pdf (reportlab, sections 1–12, formulas, tables). ANSWER/verify_solution.py prints candidate L , factorisations, divisor counts, V , $\log_{10} \Pi$, reciprocal sums, σ , abundant checks, exhaustive $L < 945$ scan, subset counts after $\sum 1/d \geq 1$, CNF/ILP dimensions, classic covering 12/12, small odd failure 57/105, BBMST mock vs naive sieveProd > 1. Residuals 0 exactly. Live sources fetched 2026-08-25: erdosproblems.com/7, /latex/7, /forum/thread/7, github.com/teorth/erdosproblems, arXiv:1811.03547/1703.02133.

References: [BBMST22] Invent. Math.230 (2022) arXiv:1811.03547+1901.11465 MR4392459; [HoNi19] Duke168 (2019) arXiv:1703.02133 MR4030365; [Sc67] Acta Arith.13 MR219515; [FFK00] Illinois44 MR1772434; [Ho15] Ann. Math.181; [Er57] Michigan4; [Site] Bloom erdosproblems.com/7; [Rafik] Zenodo18360978. AI: muse-spark-1.2-contributor-free | ROLE: @mathematician | CONTROL: OWNED | TASK: Erdos #7 investigations 1,3,4,7 $\rightarrow$ report.